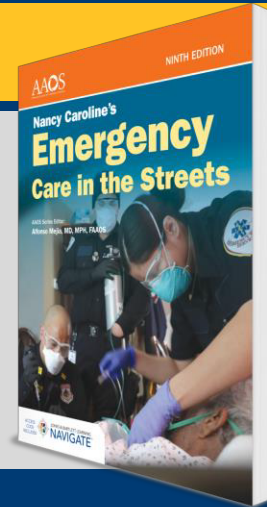


CHAPTER 20

Diseases of the Eyes, Ears, Nose, and Throat



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Chapter 20: Diseases of the Eyes, Ears, Nose, and Throat

Introduction

- Calls may involve disorders of eyes, ears, nose, and throat (EENT).
- Significant number of calls involve trauma.
- Patients may need transport to emergency department with specialists.
- Familiarity with EENT
 - Patient assessment
 - Patient education (prevention and care)

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Lecture Outline

I. Introduction

A. Paramedics may respond to calls involving disorders of the eyes, ears, nose, and throat (EENT).

1. A significant number of these calls involve trauma.
2. Familiarity with EENT conditions will help when assessing the patient.
 - a. Also allows you to educate the patient on prevention or potential care

3. Patients may need to be transported to an emergency department with access to an eye specialist or an ear, nose, and throat specialist.

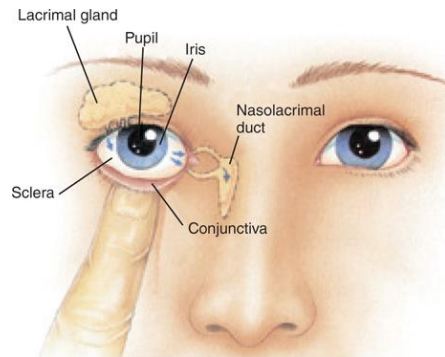
The Eye

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II. The Eye

Anatomy and Physiology Review (1 of 2)

- Oculomotor nerve (third cranial nerve)
 - Causes motion of the eyeballs and upper eyelids
 - Carries parasympathetic nerve fibers
- Optic nerve (second cranial nerve)
 - Provides sense of vision



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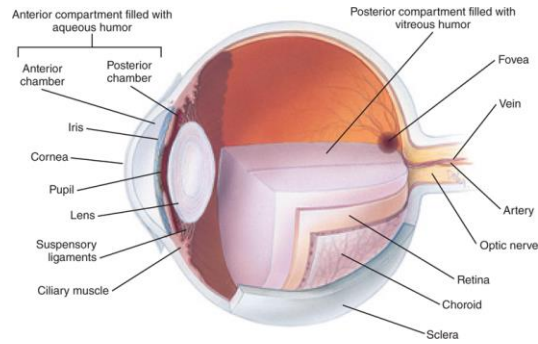
Lecture Outline

A. Anatomy and physiology review

1. Connection to the brain by two nerves
2. Oculomotor nerve (CN III; third cranial nerve)
 - a. Works with trochlear (CN IV) and abducens (CN VI) nerves to innervate the muscles that cause motion of the eyeballs and upper eyelids
 - b. Carries parasympathetic nerve fibers that cause constriction of the pupil and accommodation of the lens
3. Optic nerve (second cranial nerve)

a. Provides the sense of vision

Anatomy and Physiology Review (2 of 2)



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Patient Assessment (1 of 10)

- Ensure scene safety
- Keep patient calm
- Form general impression
- Assess airway and breathing
- Cover both eyes
- Consider pain management
- Cardiac monitoring recommended

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Lecture Outline

B. Patient assessment

1. Ensure scene safety.
2. Keep your patient calm.
3. Form a general impression.
 - a. Note environmental clues at the scene.
 - b. Note the approximate age and sex of the patient.
 - c. Note the patient's degree of distress.
4. Assess airway and breathing.

5. Early transport may improve outcomes.
6. Cover both eyes to limit damage to the affected eye through sympathetic movement.
7. Consider pain management.
8. Cardiac monitoring is recommended.
 - a. Ocular pressure can stimulate the vagus nerve.
 - b. Eye drops/medication can cause side effects such as low or high blood pressure.
9. Provide emotional care for the patient.

Patient Assessment (2 of 10)

- Obtain chief complaint and history.
- Symptoms may indicate serious ocular condition:
 - Visual loss that does not improve with blinking
 - Double vision
 - Severe eye pain
 - Foreign body sensation

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Lecture Outline

10. Obtain the chief complaint and history.
 - a. OPQRST (Onset, Provocation/palliation, Quality, Region/radiation, Severity, Timing)
 - i. How and when did symptoms begin?
 - ii. What symptoms are experienced?
 - iii. Are both eyes affected?
 - iv. Are there any underlying diseases or conditions of the eye?
11. Symptoms that may indicate a serious ocular

condition include:

- a. Visual loss that does not improve when the patient blinks
- b. Double vision
- c. Severe eye pain
- d. Foreign body sensation

Patient Assessment (3 of 10)

- Assess for:
 - Pain/tenderness
 - Swelling
 - Abnormal movement or loss of movement
 - Sensation changes
 - Circulatory changes
 - Deformity
 - Visual changes
 - Airway compromise

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Lecture Outline

12. Perform a thorough examination.
 - a. Use standard precautions.
 - b. Avoid aggravating the affected area.
 - c. Assess for:
 - i. Pain or tenderness
 - ii. Swelling
 - iii. Abnormal movement or loss of movement
 - iv. Sensation changes

- v. Circulatory changes
- vi. Deformity
- vii. Visual changes
- viii. Airway compromise

Patient Assessment (4 of 10)

- Assess structures:
 - Orbital rim
 - Ecchymosis, swelling, lacerations, tenderness
 - Eyelids
 - Ecchymosis, swelling, lacerations, abnormalities
 - Corneas
 - Foreign bodies
 - Conjunctivae
 - Redness, pus, inflammation, foreign bodies

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Lecture Outline

13. Assess the following visible ocular structures:

a. Orbital rim *Ecchymosis*

Discoloration – i. Ecchymosis, swelling, lacerations, tenderness

b. Eyelids

i. Ecchymosis, swelling, lacerations, abnormalities

c. Corneas

i. Foreign bodies

d. Conjunctivae

conjunctivae

- i. Redness, pus, inflammation, foreign bodies

Patient Assessment (5 of 10)

- Assess structures:
 - Globes
 - Redness, pigmentation, lacerations
 - Eye surface
 - Growths, discoloration, differences
 - Pupils
 - Size, shape, equality, reaction to light

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Lecture Outline

- e. Globes
 - i. Redness, abnormal pigmentation, lacerations
- f. Eye surface
 - i. Growths, discoloration, differences between eyes
- g. Pupils
 - i. PERRLA (pupils equal, round, reactive to light and accommodation)

Patient Assessment (6 of 10)

- When assessing ocular function, test:
 - Visual acuity
 - Peripheral vision
 - Ocular motility

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Lecture Outline

14. When assessing ocular function, perform the following tests:
 - a. Visual acuity
 - i. Assess ability to see large and small letters.
 - ii. Test each eye separately and document results.
 - b. Peripheral vision
 - i. Test the ability to recognize an object entering the extremes of the visual field.
 - c. Ocular motility

- i. Check the ability to move the eyes in all directions.
- ii. Check for paralysis of gaze or discoordination between the movements of the two eyes (dysconjugate gaze).

Patient Assessment (7 of 10)

- Patient may experience adverse effects:
 - Uses more than one eye medication
 - Uses too much medication
- Ask how patient administered any eye medication used.

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Lecture Outline

15. The patient may experience adverse effects if he or she:
 - a. Uses more than one eye medication
 - b. Uses too much medication
16. Ask the patient how he or she administered any medication he or she has used.
 - a. Generally recommended to wait 5 minutes between the first and second drop
 - b. 5-minute rule does not apply to emergency

situations

Patient Assessment (8 of 10)

- Eye drop uses:
 - Conjunctivitis
 - Dry, red, or itchy eyes
 - Eye pain
 - Glaucoma
 - Eye surgery
 - Herpes simplex
 - Corneal abrasions
 - Lubrication/tear production

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Lecture Outline

17. Eye drops are used for:

- a. Conjunctivitis
- b. Dry, red, or itchy eyes
- c. Eye pain
- d. Glaucoma - also
- e. Eye surgery
- f. Herpes simplex
- g. Corneal abrasions

h. Lubrication/tear production

Patient Assessment (9 of 10)

- Eye drops and lubricants applied by:
 - Gently squeezing lower eyelid
 - Applying medication
 - Having patient close eyes and roll them
 - Applying gentle pressure to corner of eyes to prevent drainage

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Lecture Outline

18. Eye drops and lubricants can be applied by:
 - a. Gently squeezing the lower eyelid to make a pouch
 - b. Applying the medication into the lower lid
 - c. Having the patient close the eyes and roll them downward
 - d. Applying gentle pressure to the corner of the eyes to prevent drainage of the medication from the eye

19. Ask patients which medications they have already taken.

Patient Assessment (10 of 10)

- Irrigation
 - May be necessary
 - Sterile water or isotonic saline solution
 - Flush from inside corner to outside
- Eye injuries
 - Should be seen in emergency department
 - May be irreversible

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Lecture Outline

20. Irrigation may be necessary for chemical or thermal burns.
 - a. Use sterile water or isotonic saline solution.
 - b. Flush liquid from the inside corner to the outside of the eye.
21. Eye injuries should be seen in the emergency department.
22. Eye injuries may be irreversible.

Contact Lenses (1 of 2)

- Only indication for removing contact lenses in prehospital setting: chemical burn in eye
- To remove hard lenses:
 - Use small suction cup
 - Moisten end with saline



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Lecture Outline

C. Patients wearing contact lenses

1. The only indication for removing contact lenses in the prehospital setting is a chemical burn of the eye.
2. To remove a hard contact lens:
 - a. Use a small suction cup, moistening the end with saline.

Contact Lenses (2 of 2)

- To remove soft lenses:
 - Place one to two drops of saline in eye
 - Gently pinch lens between your fingers and lift it off eye



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Lecture Outline

3. To remove soft lenses:

- a. Place one to two drops of saline in the eye.
- b. Gently pinch the lens between your gloved thumb and index finger, and lift it off the surface of the eye.

4. Advise emergency department staff if a patient is wearing contact lenses.

Eye Prosthesis

- Suspect if eye:
 - Does not respond to light
 - Does not move in concert with opposite eye
 - Does not appear quite like opposite eye
 - Patient indicates having one
- No harm in treating like real eye
- Strive to accurately determine eye function

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Lecture Outline

D. Patients with an eye prosthesis

1. Suspect the patient has an eye prosthesis (artificial eye) if:
 - a. The eye does not respond to light.
 - b. The eye does not move in concert with the opposite eye.
 - c. The eye does not appear quite the same as the opposite eye.
 - d. The patient says he or she has one (if you are

unsure, ask the patient).

Pathophysiology, Assessment, and Management of Specific Eye Conditions

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E. Pathophysiology, assessment, and management of specific eye conditions

Conjunctivitis

- Pink eye
- Conjunctiva becomes inflamed and red.
- Often starts in one eye and spreads to other eye
- Causes:
 - Bacteria
 - Viruses
 - Allergies
 - Foreign bodies



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Lecture Outline

1. Conjunctivitis (“pink eye”)

- a. The conjunctiva becomes inflamed and red.
- b. Most often starts in one eye and spreads to the other eye
- c. Most often caused by bacteria, viruses, allergies, chemicals, or foreign bodies
 - i. Viral conjunctivitis is often associated with an upper respiratory virus.
 - ii. Bacterial conjunctivitis is caused by bacterial

infections.

- iii. Viral and bacterial forms are highly contagious.
- iv. Allergic conjunctivitis is caused by a trigger or irritating allergen, such as pollen.
- v. Chlorine in swimming pools and air pollution are potential causes of chemical conjunctivitis.
- vi. If conjunctivitis is due to a foreign body, the eye will begin to produce tears in an attempt to flush out the object.

Conjunctivitis: Assessment and Management

- Perform general assessment of vision.
- Viral conjunctivitis resolves on its own.
- Bacterial conjunctivitis requires topical antibiotic.
- Severe allergic conjunctivitis
 - NSAIDs
 - Antihistamines
 - Topical steroid eye drops

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Lecture Outline

d. Assessment and management

- i. Perform a general assessment of the patient's vision.
- ii. Viral conjunctivitis normally resolves on its own.
- iii. Bacterial conjunctivitis requires a topical antibiotic.
- iv. Severe allergic conjunctivitis may need nonsteroidal anti-inflammatory drugs (NSAIDs), antihistamines, and topical steroid

eye drops.

Inflammation of the Eyelid (Chalazion and Hordeolum)

- Chalazion

- Swollen bump or pustule on external eyelid
- Usually painless

- Hordeolum

- Red, tender lump in eyelid or lid margin
- Internal or external (stye)



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Lecture Outline

2. Inflammation of the eyelid (chalazion and hordeolum)

- a. A protective film of oil glands and oil ducts across the eye
 - i. Chalazion: Small swollen bump or pustule on the external eyelid formed by blockage and swelling of oil gland
 - ii. Hordeolum: Infection of an oil gland that produces a red tender lump in the eyelid or at the lid margin

(a) Commonly known as a sty

Hordeolum

Inflammation of the Eyelid (Chalazion and Hordeolum): Assessment and management

- Thorough assessment of vital signs and history
- Transport to emergency department.
- Patient may be asked to treat inflammation with warm washcloth.
- Topical or oral antibiotics may be prescribed.

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Lecture Outline

b. Assessment and management

- i. A thorough assessment and transport
- ii. The patient will usually be asked to apply warm compresses for 5 to 10 minutes several times a day.
- iii. Topical or oral antibiotics may be prescribed.

Glaucoma

- Group of conditions that lead to increased intraocular pressure:
 - Open-angle glaucoma
 - Normal-tension glaucoma
 - Narrow-angle glaucoma
 - Secondary glaucoma
- Usually treated with eye drops to reduce ocular pressures

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Lecture Outline

3. Glaucoma

- a. Group of conditions that lead to increased intraocular pressure
 - i. One of the leading causes of blindness
- b. Aqueous humor
 - i. Clear, watery fluid that fills the eye's anterior chamber
 - ii. Maintains intraocular pressure, provides nutrients to the inner surface of the eye, and

helps to bend light

- iii. Circulates through the pupil and drains into the venous system by the canal of Schlemm

c. Several types of glaucoma

i. Open-angle glaucoma

- (a) Aqueous fluid drains too slowly.
- (b) Pressure builds up within the eye and damages the optic nerve.
- (c) Most common type of glaucoma

ii. Normal-tension glaucoma

- (a) Can cause vision changes with no increase in intraocular pressure

iii. Narrow-angle glaucoma (also called angle-closure glaucoma)

- (a) Aqueous fluid does not drain properly due to narrowing of the drainage channel.
- (b) Pressure builds up in the posterior chamber of the eye, which pushes the lens forward.
- (c) The lens pushes the iris into the drainage channel, completely blocking it.

d. Secondary glaucoma is a result of conditions that damage the drainage channel in the eye.

- i. Diabetes
 - ii. Eye injuries
 - iii. Leukemia
 - iv. Sickle cell anemia
 - v. Some types of arthritis
 - vi. Cataracts
- e. Incidence increases with age.
- f. Usually treated with eye drops to reduce ocular pressures

Glaucoma: Assessment and Management

- Loss of field of vision
- Rule out trauma or physical injury.
- Perform general eye assessment.
- Ophthalmologist will perform more comprehensive assessment.

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Lecture Outline

g. Assessment and management

i. Patient complaints may involve:

- (a) Loss of field of vision (specifically peripheral vision)
- (b) Tunnel vision leading up to vision loss

ii. Patients who have an acute attack of narrow-angle glaucoma may report:

- (a) Severe eye pain
- (b) Headache

- (c) Photophobia
 - (d) Nausea and vomiting
 - (e) Blurred vision
 - (f) Halos around lights
- iii. The cornea may look cloudy, and pupils often have irregular margins and can be fixed in mid-position and dilated.
 - iv. Acute narrow-angle glaucoma is a medical emergency.
 - v. Rule out trauma or physical injury.
 - vi. Perform a general eye assessment.

Use of Ophthalmoscope

- Rarely used by paramedics
- Ophthalmoscope
 - Used to examine structures of the eye
- Otoscope
 - Used to examine ear's external canal and tympanic membrane



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Central Retinal Artery Occlusion

- Blood supply to retina becomes blocked.
- Central retinal artery occlusion may cause temporary or permanent partial blindness.
- Rapid loss of vision:
 - Painless
 - Can occur during sleep
 - An emergency
 - Requires immediate transport

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Lecture Outline

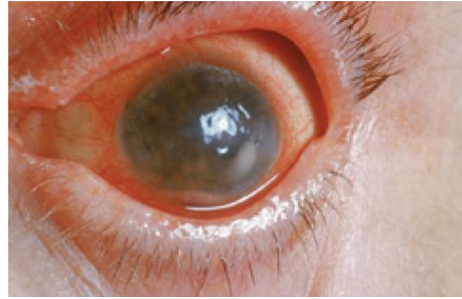
4. Central retinal artery occlusion

- a. The blood supply to the retina becomes blocked because of a clot or embolus in the central retinal artery or one of its branches.
- b. Possible causes: embolus from the carotid artery, valvular heart disease, drug abuse, fat emboli, arterial spasm, and oral contraceptive use
- c. May cause partial blindness, which may be temporary or permanent
- d. Assessment and management

- i. Sudden, painless loss of vision in one eye
- ii. Vision loss in central retinal vein occlusion may progress over 30 to 120 minutes.
- iii. Can occur during sleep
- iv. Immediate transport in a situation involving a rapid loss of vision

Iritis

- Inflammation of iris
 - Acute causes:
 - Trauma
 - Irritants
 - Chronic causes:
 - Autoimmune diseases
 - Arthritis
 - Irritable bowel disease
 - Crohn's disease



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Lecture Outline

5. Iritis

- a. Inflammation of the iris
- b. Also called anterior uveitis
- c. Third leading preventable cause of blindness
- d. Can be acute or chronic
 - i. Acute—caused by trauma or irritants and usually affects only one eye
 - ii. Chronic causes include:
 - (a) Autoimmune diseases

- (b) Different types of arthritis
 - (c) Irritable bowel disease
 - (d) Crohn's disease
- iii. Infectious causes include:
- (a) Lyme disease
 - (b) Tuberculosis TB
 - (c) Sexually transmitted diseases

Iritis: Assessment and Management

- How it presents
 - Red area surrounding iris
 - Cloudy vision
 - Unusually shaped pupil
- Focus on history
 - Acute iritis may respond to topical corticosteroids.
 - Chronic iritis should be referred to a specialist.
- Permanent disability if left untreated

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Lecture Outline

e. Assessment and management

- i. Iritis presents as a red area surrounding the iris, cloudy vision, or an unusually shaped pupil.
- ii. Focus on history.
- iii. Acute iritis usually responds well to topical corticosteroids as long as the cause is not fungal, viral, or bacterial.
- iv. Ophthalmologists are not trained to recognize iritis.

- (a) Patients should be referred to a uveitis specialist or an ocular immunologist.
- (b) Iritis can result in permanent disability if left untreated.

Papilledema (1 of 2)

- Swelling or inflammation of optic nerve
- Symptoms:
 - Headaches
 - Nausea
 - Temporary vision loss or narrowing vision fields
 - Graying in the field of vision

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Lecture Outline

Papilledema

6. Papilledema

- a. Swelling or inflammation of the optic nerve
- b. Symptoms:
 - i. Headaches
 - ii. Nausea with possible vomiting
 - iii. Temporary vision loss or narrowing vision fields
 - iv. A graying in the field of vision

Papilledema (2 of 2)

- Can be caused by:
 - Abscess
 - Tumor
 - Inner ear infection
 - Lung infection
 - Dental infection
- Other causes:
 - Meningitis
 - Fever
 - Hypertensive crisis
 - Chronic high blood pressure
 - Guillain-Barré syndrome

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Lecture Outline

- c. Can be caused by:
 - i. Abscess
 - ii. Tumor
 - iii. Inner ear infection
 - iv. Lung infection
 - v. Dental infection
 - vi. Other causes:
 - (a) Meningitis
 - (b) Fever

- (c) Hypertensive crisis
- (d) Chronic high blood pressure
- (e) Guillain-Barré syndrome

Papilledema: Assessment and Management

- Prehospital management consists of:
 - Treating symptoms
 - Transporting
 - Assessing ABCs and life threats
 - Administering analgesics or a mild sedative, if needed

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Lecture Outline

- d. Assessment and management
 - i. Diagnosis will be made by an ophthalmologist or physician.
 - ii. Prehospital management consists of:
 - (a) Treating symptoms
 - (b) Transporting
 - (c) Assessing ABCs
 - (d) Assessing for life threats
 - (e) Administering analgesics or a mild

sedative, if needed

Cellulitis of the Orbit (1 of 2)

- Periorbital cellulitis
 - More prevalent in children than adults
 - Presents as painful, red, swollen eyelid
 - Risk factors:
 - Insect bites
 - Upper respiratory disorders
 - Trauma

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Lecture Outline

7. Cellulitis of the orbit: Periorbital and orbital cellulitis

- a. Commonly caused by *Staphylococcus* and *Streptococcus* bacterial infections
- b. Periorbital cellulitis
 - i. More prevalent in children than adults
 - ii. Also known as preseptal cellulitis or eyelid cellulitis
 - iii. Presents as a painful, red, swollen eyelid

iv. Fever

v. Redness of part of the white part of the eyes

vi. Predisposed risk factors:

(a) Insect bites

(b) Upper respiratory disorders

(c) Trauma

Cellulitis of the Orbit (2 of 2)

- Orbital cellulitis
 - Medical emergency
 - Risk factors:
 - Sinusitis
 - Tooth infections
 - Ear infections
 - Trauma
 - Sinus infections

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Lecture Outline

c. Orbital cellulitis

- i. An infection within the eye socket
- ii. Considered a medical emergency
- iii. Goal of treatment: to avoid the formation of an abscess
- iv. Predisposed risk factors:
 - (a) Sinusitis
 - (b) Tooth infections
 - (c) Facial or middle ear infections

- (d) Trauma
- (e) Sinus infections

Cellulitis of the Orbit: Assessment and Management

- Treatment includes antibiotics.
- Prehospital management includes:
 - Ruling out life threats
 - Obtaining thorough history
 - Transporting to appropriate care site

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Lecture Outline

d. Assessment and management

- i. Treatment in children is usually IV antibiotics.
- ii. Adults are generally treated with oral antibiotics.
- iii. Prehospital management includes:
 - (a) Ruling out life threats
 - (b) Obtaining a thorough history
 - (c) Transporting the patient to the appropriate care site

Corneal Abrasion

- Most common eye injury
- Causes
 - Direct trauma
 - Foreign bodies
 - Contact lenses
 - Exposure to ultraviolet radiation
- Lack of prompt treatment
 - Develop ulcer
 - Blindness if left untreated

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Lecture Outline

8. Corneal abrasion or ulcer

- a. The most common eye injury seen in emergency departments
- b. Caused by direct trauma, foreign bodies, contact lenses, or exposure to ultraviolet radiation
- c. Ulcer can develop without prompt treatment of abrasion
 - i. Can cause blindness if left untreated

Corneal Abrasion: Assessment

- Symptoms include:
 - Pain
 - Redness
 - Tearing
 - Feeling of something in eye
 - Blurred vision or loss of vision
 - Photophobia
 - Headache

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Lecture Outline

d. Assessment and management

i. Symptoms include:

- (a) Pain with extraocular movement
- (b) Redness
- (c) Excessive tearing
- (d) Sensation of having something foreign in the eye
- (e) Blurred vision or loss of vision
- (f) Photophobia

(g) Headache

Corneal Abrasion: Management

- Rule out life threats.
- Take thorough history.
- Transport promptly.

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Lecture Outline

ii. Prehospital management

- (a) Rule out life threats.
- (b) Take a thorough history.
- (c) Transport promptly.

The Ears

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III. The Ears

Ears

- Primary structures for hearing and balance
- Integral to self-protection
- Disorders and injuries can leave a person unable to:
 - Communicate
 - React
 - Maintain equilibrium
- Hearing loss affects child development

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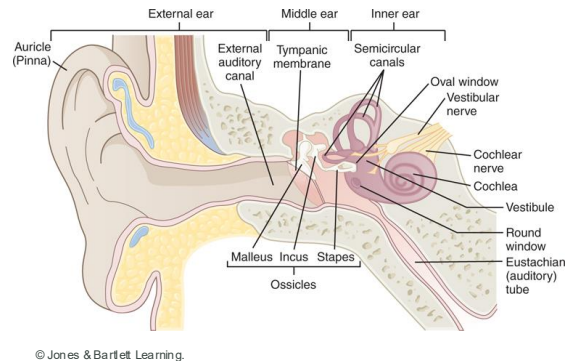
Lecture Outline

A. The primary structures for hearing and balance

1. Disorders and injuries can leave a person unable to communicate, react, and maintain equilibrium.
 - a. Changes in air pressure can cause ear discomfort.
 - b. Tumors on cranial nerves can affect the inner ear and balance, facial sensation, eye movement, facial movement, taste, and hearing.
 - c. Hearing loss can affect a child's ability to develop communication, language, and social skills.

Anatomy and Physiology Review (1 of 2)

- Divided into three anatomic parts



Lecture Outline

B. Anatomy and physiology review

1. The ear is divided into three anatomic parts.
 - a. External ear
 - i. Pinna
 - ii. External auditory canal
 - iii. External portion of the tympanic membrane (eardrum)
 - b. Middle ear

- i. Inner portion of the tympanic membrane
 - ii. Ossicles
- c. Inner ear
 - i. Cochlea
 - ii. Semicircular canals

Anatomy and Physiology Review (2 of 2)

- Sound waves travel:
 - Through ear and internal ear structures
 - From nerve impulses that travel to brain via auditory nerve
 - Brain converts these impulses into sound.

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Lecture Outline

2. Sound waves travel through the ear, and then the internal ear structures form nerve impulses that travel to the brain via the auditory nerve.
 - a. The brain converts these impulses into sound.

Patient Assessment (1 of 3)

- Observe scene for hazards.
- Assess:
 - Age and sex of the patient
 - Environmental conditions
 - Degree of distress
 - Presence of hearing aid

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Lecture Outline

C. Patient assessment

1. Possible ear injuries include the following:
 - a. Foreign objects forced into the auditory canal can damage the eardrum.
 - b. Ear infections may cause the eardrum to blister and bleed; inner ear infections may cause pressure behind the eardrum.
 - c. Blast pressure waves can burst the eardrum.
2. Observe the scene to rule out hazards to EMS

personnel and crew.

3. As you approach, assess:

- a. Approximate age and sex of the patient
- b. Environmental conditions
- c. Patient's degree of distress
- d. Whether the patient is wearing a hearing aid

Patient Assessment (2 of 3)

- Ensure ABCs and manage life threats.
- Take complete history.
- Observe ears for:
 - Drainage
 - Excess cerumen
 - Inflammation
 - Swelling

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Lecture Outline

4. Ensure airway patency, breathing adequacy, and circulation.
5. Manage life threats.
6. Take a complete history.
7. Observe ears for:
 - a. Drainage
 - b. Excess cerumen (earwax)
 - c. Inflammation
 - d. Swelling

Patient Assessment (3 of 3)

- Have patient rate pain.
- Ask about:
 - Changes in hearing
 - Tinnitus
 - Dizziness
- Inspect for:
 - Wounds
 - Swelling
 - Drainage
 - Battle sign of mastoid process

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Lecture Outline

8. Have the patient rate his or her pain using OPQRST.
9. Ask the patient about:
 - a. Changes in hearing
 - b. Tinnitus (ringing in the ears)
 - c. Dizziness
10. Inspect and palpate for:
 - a. Wounds
 - b. Swelling

c. Drainage

d. Battle sign of the mastoid process of the skull

Pathophysiology, Assessment, and Management of Specific Ear Conditions

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D. Pathophysiology, assessment, and management of specific ear conditions

Impacted Cerumen (1 of 2)

- Yellowish oily substance found in ear canal
- May present as:
 - Wet: sticky brown color
 - Dry: grayish flaky substance
- Can become impacted and cause pressure on eardrum

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Lecture Outline

1. Impacted cerumen

- a. Cerumen is yellowish, oily substance found in the outer ear canal (earwax)
 - i. Helps prevent dirt and water from entering the middle ear canal and may protect the ear from bacteria or fungus
 - ii. May present as:
 - (a) Wet—a sticky brown color
 - (b) Dry—a grayish flaky substance

- iii. Can become impacted and cause pressure against the eardrum
- b. More common in older adults

Impacted Cerumen (2 of 2)

- Risk factors include:
 - Abnormal ear canal shape
 - Diseases that cause increased cerumen
 - Improper use of cotton swabs

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Lecture Outline

- c. Other risk factors:
 - i. Abnormal ear canal shape
 - ii. Diseases that cause increased production of cerumen
 - iii. Improper use of cotton swabs

Impacted Cerumen: Assessment and Management

- Symptoms may include:
 - Pressure or fullness in ears
 - Ringing in ears
 - Loss of hearing
 - Pain or itching
- Prehospital treatment
 - Thorough history
 - Visual inspection of ear

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Lecture Outline

d. Assessment and management

i. Symptoms:

- (a) Sensation of pressure or fullness in the ears
- (b) Dizziness
- (c) Ringing in the ears
- (d) Loss of hearing
- (e) Pain or itching in the ears

e. Prehospital treatment includes:

- i. Thorough history
 - ii. Visual inspection of the ear
- f. Treatment is aimed at removing the excess cerumen.
 - i. Do not attempt to extract the material yourself.
 - ii. If left untreated, infection and irritation can occur.

Labyrinthitis

- Feeling of vertigo or loss of balance after an ear infection or upper respiratory infection
- Effect on nerves from irritation and swelling
- Symptoms may include:
 - Loss of balance
 - Ringing in ears
 - Loss of hearing
 - Vomiting
- Possible permanent hearing loss

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Lecture Outline

2. Labyrinthitis

- a. Most commonly recognized as the feeling of vertigo or loss of balance after an ear infection or upper respiratory infection
- b. Effect on the nerves of the inner ear and a loss of balance from irritation and swelling of the inner ear
- c. Other symptoms:
 - i. Ringing in the ears

- ii. Dizziness
- iii. Loss of hearing
- iv. Nausea
- v. Vomiting

Labyrinthitis: Assessment and Management (1 of 2)

- Prehospital treatment directed at:
 - Reducing nausea and vomiting
 - Transporting patient in position of comfort
- Rule out serious disorders with CT scan and MRI.

labyrinthitis

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Lecture Outline

d. Assessment and management

i. Prehospital treatment is directed at:

- (a) Reducing the severity of the nausea and vomiting
- (b) Transporting the patient in a position of comfort

ii. Serious disorders will need to be ruled out by a CT scan and an MRI.

Labyrinthitis: Assessment and Management (2 of 2)

- Prompt treatment of respiratory and ear infections
- Hospital treatment:
 - Antiemetic for nausea and vomiting
 - Antihistamine to reduce swelling
 - Antivertigo medicine
 - Diazepam (Valium) as a sedative/muscle relaxant

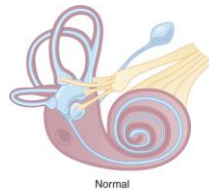
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Lecture Outline

- iii. Hospital treatment includes:
 - (a) Antiemetic for nausea and vomiting
 - (b) Antihistamine for swelling
 - (c) Antivertigo medicine
 - (d) Diazepam as sedative

Meniere Disease

- Chronic condition
- Characterized by:
 - Dizziness described as spinning vertigo
 - Low-frequency hearing loss
 - Tinnitus
 - Feeling of fullness in affected ear
- Overproduction and defective absorption of endolymphatic fluid



Normal



Ménière disease

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Lecture Outline

3. Meniere disease

- a. Chronic condition of the inner ear characterized by:
 - i. Dizziness described as spinning vertigo
 - ii. Low-frequency hearing loss
 - iii. Tinnitus
 - iv. Feeling of fullness in the affected ear
- b. Overproduction and defective absorption of endolymphatic fluid, which increases the volume

and pressure within the labyrinth of the inner ear

- i. Distention results in rupture and mixing of the endolymph and perilymph fluids.
 - ii. This mixture disrupts the balance of fluid and electrolytes within the labyrinth and damages the vestibular and cochlear hair cells.
- c. Attacks of less than 2 hours in the early stages; altered balance up to 2 days
- i. As the disease progresses, symptoms last hours to days.
 - ii. Permanent tinnitus, moderate to severe hearing loss, and chronic unsteadiness may result.

Meniere Disease: Assessment and Management

- Prehospital care
 - Treating nausea and vomiting with antiemetic
- Physician
 - May treat with diuretics and an antiemetic
 - Limited success with surgical procedures

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Lecture Outline

d. Assessment and management

- i. Prehospital care includes treating the nausea and vomiting with an antiemetic.
- ii. The physician may treat with diuretics and an antiemetic.

Otitis Externa and Media

- Infection resulting from bacterial growth in ear canal
 - Otitis externa
 - Otitis media
 - More common in children than adults

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Lecture Outline

4. Otitis externa and media

- a. Infection that results from bacterial growth in the ear canal
 - i. Otitis externa—infection of the outer ear
 - ii. Otitis media—infection of the middle ear
- b. More common in children than adults
- c. Most commonly bacterial infections
 - i. Otitis externa can also be an allergic or fungal reaction.

- ii. Otitis media can be virally induced.

Otitis Externa and Media: Assessment and Management

- Signs and symptoms may include:
 - Edema and erythema
 - Diminished hearing acuity
 - Inflamed, bulging tympanic membrane
- Prehospital treatment includes relieving unbearable symptoms.

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Lecture Outline

d. Assessment and management

i. Signs and symptoms:

- (a) Pain
- (b) Itching
- (c) Edema and erythema
- (d) Diminished hearing acuity
- (e) Inflamed, bulging tympanic membrane on exam with otoscope

ii. Prehospital treatment should be directed at

relieving unbearable symptoms.

- (a) Monitor the patient's condition.
- (b) Administer pain medication when necessary.

The Nose

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IV. The Nose

Nose (1 of 2)

- Susceptible to injury because of its prominence
- Allergens, particles, and chemicals can cause inflammation, infection, and injury.
- Inside is extremely vascular:
 - Excellent route for some medicines
 - Nasal tissue can bleed
 - Drug abuse
- Nasal mucosa offers short route to brain.

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Lecture Outline

A. Susceptible to injury because of prominent location on the face

1. The nose acts as a filter, humidifier, and heater for air that enters the body.
2. Allergens, particles, and chemicals can cause inflammation, infection, and injury.
 - a. Complications from nasal disorders are common.
 - b. Many symptoms begin as nasal infections.
3. The inside of the nose is extremely vascular.

- a. Excellent route for some medicines
- b. Can cause nasal tissue to bleed
- c. Nasal mucosa offers a short route to the brain.

Nose (2 of 2)

- Smelling disorders
 - Anosmia (total loss of sense of smell)
 - Dysosmia (distorted sense of smell)
 - Hyperosmia (increased sensitivity to smell)
 - Hyposmia (decreased sense of smell)
 - Presbyosmia (loss of smell from normal aging)
 - Affect sense of taste

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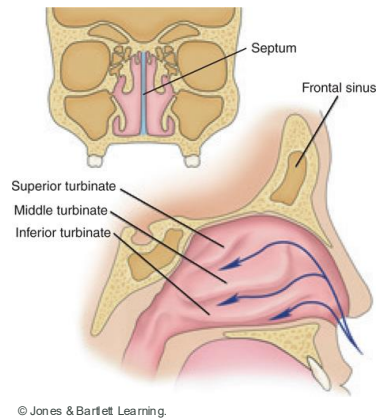
Lecture Outline

4. Loss of smelling sensation has many causes.
 - a. Aging, smoking, allergies, rhinitis, polyps, flu, coronavirus disease 2019 (COVID-19), medications, traumatic brain injury
 - b. Smelling disorders
 - i. Anosmia (total loss of sense of smell)
 - ii. Dysosmia (distorted sense of smell)
 - iii. Hyperosmia (increased sensitivity to smell)

- iv. Hyposmia (decreased sense of smell)
- v. Presbyosmia (loss of smell from normal aging)

Anatomy and Physiology Review (1 of 2)

- One of two primary entry points for oxygen
- Warms and humidifies air as it enters the body



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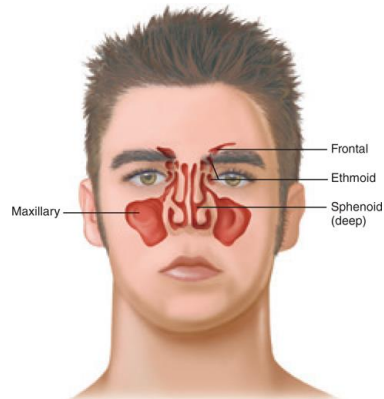
Lecture Outline

B. Anatomy and physiology review

1. One of two primary entry points for oxygen
2. Warms and humidifies air

Anatomy and Physiology Review (2 of 2)

- Contains bony structures
- Connected with sinuses



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Lecture Outline

3. Contains bony structures
4. Connected with the sinuses

Patient Assessment (1 of 2)

- Look for environmental clues.
- Ensure scene safety.
- Assess airway and breathing.
- Determine level of distress.
- Nasal cavities are prone to bleeding.

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Lecture Outline

C. Patient assessment

1. Look for environmental clues.
2. Ensure that the scene is safe.
3. Determine whether airway and breathing are sufficient.
4. Determine the patient's level of distress.
5. The vascular nature of the nasal cavities makes them susceptible to bleeding.

Patient Assessment (2 of 2)

- Insert airway adjunct as needed.
- Do not insert a nasopharyngeal airway or attempt nasotracheal intubation with:
 - Suspected nasal fractures
 - CSF or blood leakage from the nose
- Inquire about patient's history of nose conditions.

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Lecture Outline

6. Insert an airway adjunct as needed.
 - a. Do not insert a nasopharyngeal airway or attempt nasotracheal intubation in any patient with:
 - i. Suspected nasal fractures
 - ii. CSF or blood leakage from the nose
7. Inquire about a previous history of nose conditions or bleeding.
8. Always consider a hypertensive crisis when an older person has a nosebleed.

Pathophysiology, Assessment, and Management of Specific Nose Conditions

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D. Pathophysiology, assessment, and management of specific nose conditions

Epistaxis (1 of 2)

- Nosebleed
- Causes:
 - Digital trauma
 - Dryness
 - Hypertension

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Lecture Outline

1. Epistaxis

a. Nosebleed

b. Most common cause is digital trauma

c. Other causes include:

i. Dryness

ii. Hypertension

Epistaxis (2 of 2)

- Anterior
 - Bleed fairly slowly
 - Self-limiting and resolve quickly
- Posterior
 - More severe
 - Often cause blood to drain into throat
 - Causes nausea and vomiting

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Lecture Outline

d. Two types

i. Anterior

(a) Usually self-limiting and resolve quickly

ii. Posterior

(a) Usually more severe

(b) Often cause blood to drain into the patient's throat, causing nausea and vomiting

Epistaxis: Assessment and Management

- Try to estimate blood loss.
- Place non-trauma patient in sitting position and pinch nostrils together.
- Direct patient not to sniff or blow his or her nose.



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Lecture Outline

e. Assessment and management

- i. Try to estimate the amount of blood loss.
- ii. Place a non-trauma patient in a sitting position, leaning forward, and pinch his or her nostrils together with firm pressure for 20 minutes.
- ii. Direct the patient not to sniff or blow his or her nose.

Foreign Body

- Mostly seen in pediatric patients
- Pressure in nasal passage can cause:
 - Tissue necrosis
 - Inflammation
 - Swelling
- Inflammation can cause tissue ulceration and epistaxis.
- Nasal blockage can lead to sinusitis.

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Lecture Outline

2. Foreign body

- a. Most likely to be seen in pediatric patients
- b. Pressure in the nasal passage can cause:
 - i. Tissue necrosis
 - ii. Inflammation
 - iii. Swelling
- c. Tissue ulceration and epistaxis caused by inflammation
- d. Sinusitis caused by nasal blockage

Foreign Body: Assessment and Management

- Determine if condition is life-threatening.
- Persistent, foul-smelling, purulent discharge:
 - Suspect foreign body
 - Let it drain
- Transport patient in position of comfort.

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Lecture Outline

e. Assessment and management

- i. Determine if the foreign body presents a life-threatening condition.
- ii. Any persistent, foul-smelling, purulent discharge from the nares should lead to suspicion of a foreign body.
 - (a) Let it drain.
- iii. Transport the patient in a position of comfort.

- (a) Limit the ability of gravity to introduce the object further into the cavity.
 - (b) Prevent aspiration.
- iv. Pain management or sedation may be necessary.

Rhinitis

- Inflammation of nasal cavity
- May be caused by:
 - Bacterial infection
 - Viral infection
 - Allergens
 - Medications
 - Changes in environmental temperature
 - Other factors

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Lecture Outline

3. Rhinitis

- a. Inflammation of the nasal cavity
- b. May be caused by bacterial or viral infection, allergens, medications, or changes in environmental temperature
- c. Can also be caused by:
 - i. Certain medications
 - ii. Foreign bodies
 - iii. Irritants in the air

iv. Hormonal changes in pregnancy

Rhinitis: Assessment and Management

- Signs and symptoms may include:
 - Nasal congestion
 - Sneezing
 - Itchy, runny nose and eyes
 - Postnasal drip
- Keep patient in Fowler position.

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Lecture Outline

d. Assessment and management

i. Signs and symptoms include:

- (a) Nasal congestion
- (b) Sneezing
- (c) Itchy runny nose
- (d) Itchy eyes
- (e) Postnasal drip
- (f) Cough

- ii. Keep the patient in the Fowler position.
- iii. Provide transport.

Sinusitis

- Sinus inflammation
- Drainage from sinuses becomes disrupted.
- Variety of symptoms
- Affects 28.9 million adult Americans per year
- More common in young children and older adults

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Lecture Outline

4. Sinusitis

- a. Sinus inflammation
- b. Occurs when drainage from the sinuses becomes disrupted
- c. Facial pressure and pain, sore throat, nasal congestion, toothache, headache, fever, chills, and muscle aches and pains
- d. Affects 28.9 million adult Americans per year according to the CDC

- e. Young children and older adults are more susceptible.

Sinusitis: Assessment and Management

- Condition can be chronic, acute, or recurrent.
- Treat any respiratory compromise and transport.
- Treatment:
 - Reducing inflammation
 - Draining sinuses

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Lecture Outline

- f. Assessment and management
 - i. The condition can be chronic, acute, or recurrent.
 - ii. Prehospital management should include treatment of any respiratory compromise and transport.
 - iii. Treatment is aimed at reducing inflammation and draining the sinuses.
 - (a) Mild to moderate symptoms can be treated with a saline rinse and decongestant.

- (b) Antibiotics are typically prescribed after 7 to 10 days.

The Throat

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V. The Throat

Throat (1 of 2)

- Disorders of pharynx and larynx:
 - Inflammation
 - Infections
 - Abnormal growths
- Infections common in children
- Problems can be exacerbated by swallowing problems.

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Lecture Outline

A. Acute inflammation and infections, chronic inflammation, or abnormal growths

1. Specific disorders include:

- a. Vocal cord polyps and nodules
- b. Contact ulcers
- c. Vocal cord paralysis
- d. Laryngoceles
- e. Laryngeal papillomas

f. Cancer

2. Throat infections are common in children.
3. Throat problems can be exacerbated by swallowing problems.
 - a. Cranial nerves VI, VII, IX, and XII all play a role in swallowing.
 - i. Neurologic problems associated with stroke or trauma can cause swallowing difficulty.
 - b. Aspiration pneumonia is a life-threatening condition.

Throat (2 of 2)

- Esophageal disorders can affect throat.
- Esophageal reflux:
 - Valve only partially closes or opens too much.
 - Symptoms include:
 - Burning sensation in chest
 - Indigestion
 - Change in voice tone

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Lecture Outline

4. Esophageal disorders can affect the throat.
 - a. The valve at the end of the esophagus keeps acidic stomach contents from coming back up the throat.
 - b. In esophageal reflux, the valve only partially closes or opens too much.
 - i. Symptoms include:
 - (a) Burning sensation in the chest
 - (b) Indigestion

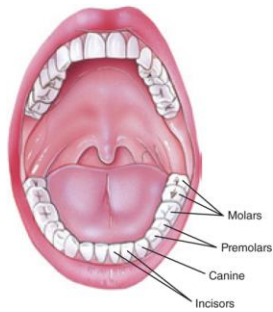
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(c) Change in voice tone

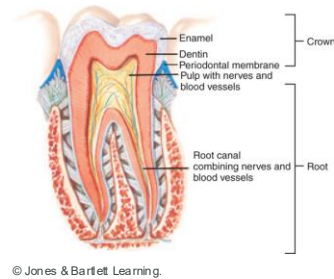
ii. Can cause a precancerous condition

Anatomy and Physiology Review (1 of 2)

- Assessment begins at opening of mouth with teeth.



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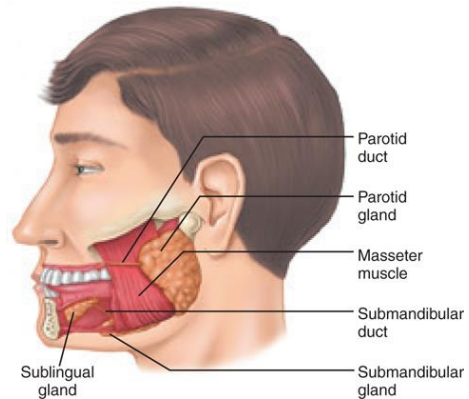
Lecture Outline

B. Anatomy and physiology review

- Assessment begins at the opening of the mouth with the teeth.

Anatomy and Physiology Review (2 of 2)

- Nerves:
 - Hypoglossal
 - Glossopharyngeal
 - Trigeminal
 - Facial



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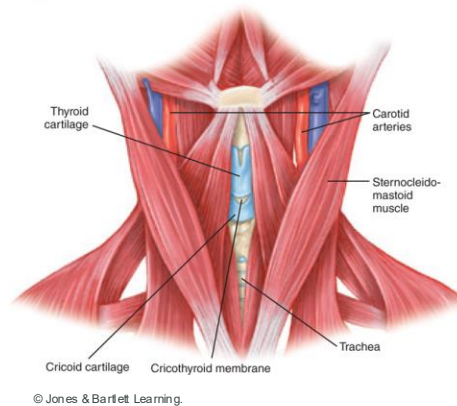
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Lecture Outline

2. Hypoglossal, glossopharyngeal, trigeminal, and facial nerves supply the mouth and its structures.

The Neck (1 of 3)

- Anterior
 - Thyroid and cricoid cartilage
 - Trachea
 - Muscles and nerves



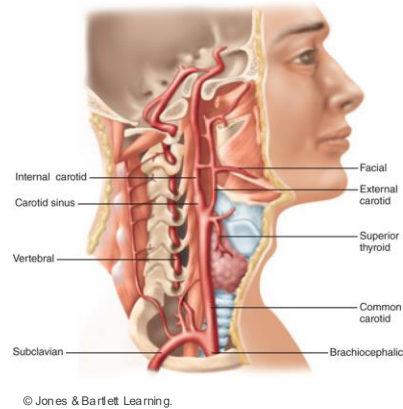
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Lecture Outline

3. The neck consists of the anterior and posterior portions.
 - a. The anterior part of the neck includes:
 - i. Thyroid and cricoid cartilage
 - ii. Trachea
 - iii. Numerous muscles and nerves
 - iv. Major blood vessels

The Neck (2 of 3)

- Major blood vessels
 - Internal and external carotid arteries
 - Internal and external jugular veins

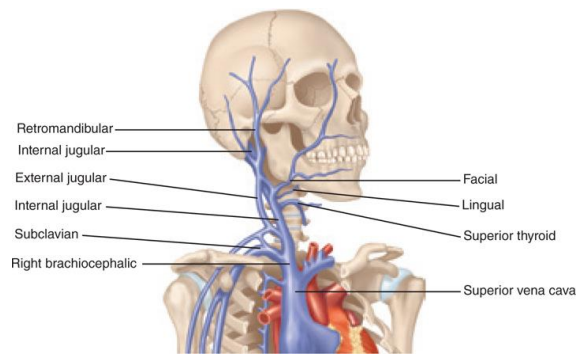


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Lecture Outline

- The major blood vessels in this area are the internal and external carotid arteries.
- Vertebral arteries run laterally to the cervical vertebrae in the posterior part of the neck.

The Neck (3 of 3)



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Lecture Outline

C. Patient assessment

1. Patients with swallowing abnormalities or copious mucous production should be placed in a position to allow drainage.
 - a. Lateral recumbent or recovery position
2. Assessing stroke patients must include early recognition of airway threats.
3. For patients who cannot protect their airways and are at risk for aspiration into the lungs, intubation

should be considered.

4. In your assessments, you should consider epiglottitis if there are symptoms of:

- a. Sore throat
- b. Drooling
- c. A forward-hanging head

Patient Assessment

- Swallowing abnormalities
 - Require position to allow drainage
 - Lateral recumbent or recovery position
- Assessments should consider epiglottitis with symptoms of:
 - Sore throat
 - Drooling
 - A forward-hanging head

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Lecture Outline

C. Patient assessment

1. Patients with swallowing abnormalities or copious mucous production should be placed in a position to allow drainage.
 - a. Lateral recumbent or recovery position
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Pathophysiology, Assessment, and Management of Specific Throat Conditions

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D. Pathophysiology, assessment, and management of specific throat conditions

Dentalgia and Dental Abscess (1 of 2)

- Dentalgia
 - Toothache
 - Starting point of dental abscess
 - Cavity harbors bacteria

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Lecture Outline

1. Dentalgia and dental abscess

- a. Dentalgia (toothache) can be the start of a dental abscess.

Dentalgia

Dentalgia and Dental Abscess (2 of 2)

- Dental abscess
 - Bacteria growth spreads into gums, facial tissue, bones, and/or neck
 - Pain relieved when ruptured
 - Drains pus
 - Reduces swelling



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Lecture Outline

- i. A dental abscess occurs when bacteria growth spreads directly from a cavity into the gums, facial tissue, bones, and/or neck.
- ii. It may have to be drained surgically.

Dentalgia and Dental Abscess: Assessment and Management

- Infection may become systemic.
- Abscess in throat, neck, or under tongue can affect patient's ability to breathe.
- Prehospital treatment:
 - Aimed at relieving symptoms
 - Rinse with warm water if ruptured.

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Lecture Outline

b. Assessment and management

- i. Infection may have become systemic if the patient has:
 - (a) Fever
 - (b) Chills
 - (c) Nausea
 - (d) Vomiting
- ii. An abscess in the throat, neck, or under the tongue can affect the ability to breathe.

- iii. Prehospital treatment is aimed at relieving the symptoms.
 - (a) Drainage into the mouth should be rinsed with warm water.
- iv. Encourage transport.

Diseases of Oral Soft Tissue

- Can be root cause of other health problems
- Include:
 - Cold sores
 - Canker sores
 - Thrush
 - Leukoplakia
 - Gingivitis
 - Bad breath

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Lecture Outline

2. Diseases of oral soft tissue

- a. Can be root cause of other health problems
 - i. Gum disease has been linked to heart disease, stroke, diabetes, osteoporosis, and low-birth-weight babies.
- b. Some common mouth disorders:
 - i. Cold sores
 - ii. Canker sores
 - iii. Oral candidiasis (thrush)

- (a) Yeast infection that causes white patches in the mouth or on the oral mucosa

iv. Leukoplakia

- (a) Causes excess cell growth in mouth, cheek, or gums
- (b) Presents as white patches

v. Gingivitis

vi. Bad breath

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Diseases of Oral Soft Tissue: Assessment and Management

- Patient may be embarrassed.
- Assess lumps and sores in mouth.
- Rule out urticaria and allergic reactions.

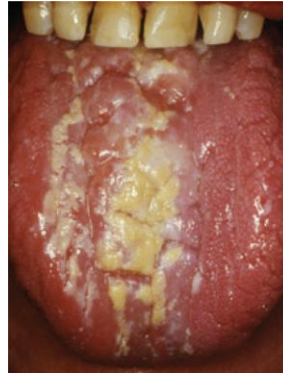
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Lecture Outline

- c. Assessment and management
 - i. Be aware that patients may be too embarrassed to discuss condition.
 - ii. Rule out urticaria and allergic reactions when assessing mouth sores or lumps.

Oral Candidiasis (Thrush)

- *Candida albicans* fungus accumulates on mouth lining.
- Patient will have creamy white lesions.
- Lesions spread:
 - Roof of mouth
 - Gums
 - Tonsils
 - Pharynx



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Lecture Outline

3. Oral candidiasis

- a. Condition in which the fungus *Candida albicans* fungus accumulates on the lining of the mouth
- b. Creamy white lesions on the tongue and inner cheeks
 - i. May be painful and may bleed as they are rubbed or scraped
 - ii. Can spread to the roof of the mouth, gums, tonsils, or pharynx

Oral Candidiasis: Assessment and Management (1 of 3)

- Most likely to be found in:
 - Babies
 - Patients with compromised immune systems
 - Patients who wear dentures
 - Patients who use inhaled corticosteroids

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Lecture Outline

c. Assessment and management

i. Thrush is most likely to be found in:

- (a) Infants
- (b) Patients with compromised immune systems
- (c) Patients who wear dentures
- (d) Patients who use inhaled corticosteroids

Oral Candidiasis: Assessment and Management (2 of 3)

- Symptoms include:
 - Bleeding
 - Pain
 - Loss of taste
 - Cracking and redness at corners of mouth
 - Cottony feeling in mouth
 - Sensation of food stuck in throat

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Lecture Outline

ii. Additional symptoms include:

- (a) Bleeding
- (b) Pain
- (c) Cracking and redness at the corners of the mouth
- (d) Loss of taste
- (e) Cottony feeling in the mouth
- (f) In severe cases, lesions can move down the esophagus, causing the sensation that food is getting stuck in the throat when

swallowing.

Oral Candidiasis: Assessment and Management (3 of 3)

- Patients at increased risk:
 - HIV/AIDS
 - Cancer
 - Diabetes
 - Vaginal yeast infections
- Treat higher priorities.
- Make patient comfortable.

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Lecture Outline

- iii. Patients at increased risk have a medical history of:
 - (a) HIV/AIDS
 - (b) Cancer
 - (c) Diabetes
 - (d) Vaginal yeast infections
- iv. Treat higher priorities.
- v. Make the patient comfortable and encourage follow-up with physician.

Ludwig Angina

- Type of cellulitis
- Caused by bacteria from an infected tooth root or mouth injury
- Occurs on floor of mouth or under tongue
- Physical exam may show redness and swelling of neck or under chin.

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Lecture Outline

4. Ludwig angina

- a. Type of cellulitis caused by bacteria from an infected tooth root or mouth injury
- b. Occurs on the floor of the mouth under the tongue
- c. Rapid swelling and airway obstruction
- d. Redness and swelling of the neck or under the chin
 - i. Tongue may be swollen.
- e. An airway through the nasal passages possibly

needed

Ludwig Angina: Assessment and Management

- Symptoms may include:
 - Difficulty breathing or swallowing
 - Neck pain and swelling
 - Fever
 - Drooling
 - Altered speech sounds
- Prehospital treatment
 - Early treatment with steroids to reduce swelling
 - Requires aggressive management of airway in severe cases
 - Careful attention to condition and smells from mouth

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Lecture Outline

f. Assessment and management

i. Symptoms may include:

- (a) Difficulty breathing
- (b) Difficulty swallowing
- (c) Neck pain
- (d) Neck swelling
- (e) Fever
- (f) Drooling

- (g) Altered speech sounds
- ii. Prehospital treatment requires aggressive management of the airway in severe cases.

Epiglottitis

- Inflammation of epiglottis
 - Blocks trachea
 - Obstructs airway
 - Often result of *Haemophilus influenzae* type b virus
- Hib vaccine

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Lecture Outline

5. Epiglottitis

- a. Inflammation of the epiglottis
 - i. Blocks the trachea and obstructs the airway
 - ii. Often result of the *Haemophilus influenzae* type b virus in unvaccinated adults

Epiglottitis: Assessment and Management (1 of 2)

- Symptoms
 - Fever
 - Sore throat
 - Painful swallowing
 - Stridor
- Signs
 - Sick and anxious
 - Tripod position or sniffing position
 - Increased work of breathing
 - Pallor or cyanosis

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Lecture Outline

b. Assessment and management

i. Symptoms include:

- (a) Fever
- (b) Sore throat
- (c) Painful swallowing
- (d) Stridor
- (e) Respiratory distress

ii. Signs include:

- (a) Patient will look sick and anxious.

Epiglottitis

- (b) Patient will sit upright in the classic tripod position or in the sniffing position.
- (c) Patient may be drooling.
- (d) Work of breathing is increased.
- (e) Pallor or cyanosis may be evident.

Epiglottitis: Assessment and Management (2 of 2)

- Transport to appropriate hospital.
 - Minimize on-scene time.
 - Do not agitate patient.
 - Do not attempt to look in mouth.
 - Alert receiving personnel.

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Lecture Outline

- iii. Transport to an appropriate hospital while maintaining the airway.
 - (a) Minimize on-scene time.
 - (b) Do not attempt procedures that might agitate the patient.
 - (c) Do not attempt to look in the mouth.
 - (d) Alert receiving personnel of suspected diagnosis and patient's condition.

Laryngitis

- Swelling and inflammation of larynx
- Causes:
 - Overuse
 - Pneumonia
 - Irritants and chemicals
 - Bronchitis
 - Allergies

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Lecture Outline

6. Laryngitis

- a. Swelling and inflammation of larynx associated with hoarseness or loss of voice
- b. Can be the result of overuse
- c. Most common form caused by a virus
 - i. Can also be caused by:
 - (a) Pneumonia
 - (b) Irritants
 - (c) Chemicals

- (d) Gastroesophageal reflux disease
- (e) Bronchitis
- (f) Allergies
- (g) Bacterial infections

Laryngitis: Assessment and Management

- Symptoms:
 - Fever
 - Hoarseness
 - Swollen lymph nodes or glands
- Consider a fracture of hyoid bone.
- Have patient follow up with physician.

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Lecture Outline

d. Assessment and management

i. Symptoms include:

- (a) Fever
- (b) Hoarseness
- (c) Swollen lymph nodes or glands in the neck

ii. Obtain a good history to rule out evolving upper airway obstruction or an allergic reaction.

- iii. Consider fracture of the hyoid bone.
- iv. Have the patient follow up with a physician.

Tracheitis

- Bacterial infection of trachea
- Caused by *Staphylococcus aureus*
- Frequently occurs in children following upper respiratory infection
- Trachea easily blocked by swelling
- Can be life threatening

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Lecture Outline

7. Tracheitis

- a. Bacterial infection of the trachea caused by *Staphylococcus aureus*
- b. Frequently occurs in young children following a viral upper respiratory infection
 - i. The trachea is easily blocked by swelling.
 - ii. It can be a life-threatening condition.

Tracheitis: Assessment and Management (1 of 2)

- Symptoms:
 - Croup-like cough
 - Difficulty breathing
 - High fever
- Signs:
 - Tripod positioning
 - Intercostal retractions

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Lecture Outline

c. Assessment and management

i. Symptoms include:

- (a) Deep croup-like cough
- (b) Difficulty breathing
- (c) High fever
- (d) High-pitched stridor with breathing

ii. Patients may exhibit:

- (a) Tripod positioning

- (b) Intercostal retractions
- (c) Can proceed from respiratory distress to respiratory failure if not addressed

Tracheitis: Assessment and Management (2 of 2)

- Prehospital care is supportive:
 - Minimize stress to patient.
 - Administer 100% oxygen.
 - Use pulse oximetry.
 - Monitor vital signs.
 - Be prepared for difficult intubation.
 - Transport promptly.

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Lecture Outline

- iii. Supportive prehospital care
 - (a) Minimize stress to the patient.
 - (b) Administer 100% oxygen.
 - (c) Use pulse oximetry.
 - (d) Monitor vital signs.
 - (e) Be prepared for difficult intubation.
 - (f) Transport promptly.

Tonsillitis

- Swelling and inflammation of tonsils
- Causes:
 - Usually, viral infections
 - Bacterial infections



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Lecture Outline

8. Tonsillitis

- a. Swelling and inflammation of the tonsils
- b. Usually caused by viral infections, but can also be caused by bacteria

Tonsillitis: Assessment and Management (1 of 2)

- Symptoms:
 - Swollen tonsils
 - Sore throat
 - Difficulty swallowing
- Signs:
 - Red, swollen tonsils
 - White or yellow coating or patches
 - Fever
 - Sore throat

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Lecture Outline

c. Assessment and management

i. Symptoms include:

- (a) Swollen tonsils
- (b) Sore throat
- (c) Difficulty swallowing

ii. Patients will present with the following:

- (a) Red, swollen tonsils
- (b) White or yellow coating or patches on the tonsils

- (c) Fever
- (d) Sore throat

Tonsillitis: Assessment and Management (2 of 2)

- Other signs:
 - Pain when swallowing
 - Enlarged or tender lymph nodes in neck
 - Bad breath
 - Headache
 - Stiff neck
 - Drooling
- Transport to ED for evaluation.

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Lecture Outline

- (e) One or more of the following:
 - (1) Pain when swallowing
 - (2) Enlarged or tender lymph nodes in the neck
 - (3) Bad breath
 - (4) Headache
 - (5) Stiff neck
 - (6) Drooling
- iii. Transport for further evaluation.

Pharyngitis

- Inflammation of pharynx
- Often due to rapid onset of sore throat
- No discomfort or pain with swallowing



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Lecture Outline

9. Pharyngitis

- a. Inflammation of the pharynx
- b. Often due to a rapid onset of sore throat without discomfort or pain with swallowing

pharyngitis

Pharyngitis: Assessment and Management

- Symptoms may include:
 - Fever
 - Headache
 - Pharyngeal erythema
 - Purulent, patchy yellow, gray, or white exudate
 - Ulcers on soft palate
- Assess for airway obstruction.
- Follow-up with emergency department

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Lecture Outline

c. Assessment and management

i. Symptoms include:

- (a) Fever
- (b) Pharyngeal erythema
- (c) Headache
- (d) Purulent, patchy yellow, gray, or white exudate
- (e) Nasal congestion
- (f) Hoarseness

(g) Cough

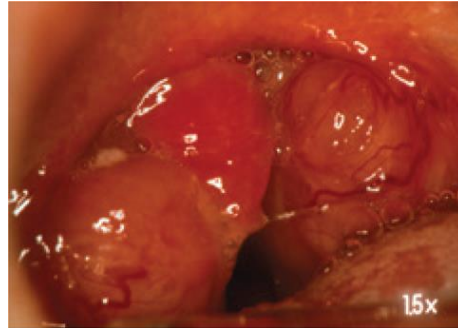
(h) Ulcers on the soft palate

ii. Treatment involves follow-up with the emergency department.

iii. A major prehospital concern is assessment for partial airway obstruction.

Peritonsillar Abscess

- Collection of infected material around tonsils
- Complication of tonsillitis
- From bacterial infection
- Rare thanks to antibiotics



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Lecture Outline

10. Peritonsillar abscess

- a. Collection of infected material around the tonsils
- b. Complication of tonsillitis made rare thanks to use of antibiotics

Peritonsillar Abscess: Assessment and Management (1 of 2)

- Infection in:
 - One or both tonsils
 - Roof of mouth
 - Neck or chest
- Symptoms:
 - Chills
 - Difficulty and pain opening mouth
 - Facial swelling
 - Fever
 - Drooling or inability to swallow saliva
 - Tender glands of the jaw and throat

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Lecture Outline

c. Assessment and management

- i. One or both tonsils are infected.
- ii. Roof of the mouth and neck or chest may be infected.
- iii. Patient may have:
 - (a) Chills
 - (b) Difficulty opening the mouth
 - (c) Pain with opening the mouth
 - (d) Facial swelling

- (e) Fever
- (f) Drooling or inability to swallow saliva
- (g) Headache
- (h) Muffled voice
- (i) Sore throat
- (j) Tender glands of the jaw and throat

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Peritonsillar Abscess: Assessment and Management (2 of 2)

- Treatment:
 - Involves antibiotics and draining abscess
 - May include tonsillectomy
 - Transport patient to hospital

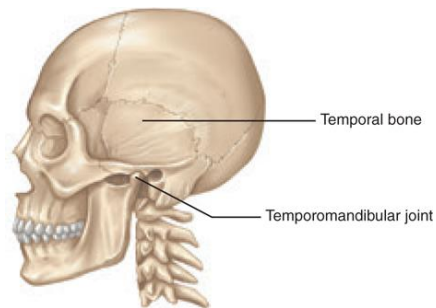
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Lecture Outline

- iv. Treatment includes antibiotics and draining the abscess
 - (a) May include tonsillectomy
 - (b) Hospital transport

Temporomandibular Joint Disorders

- TMJ disorders
- Causes include:
 - Arthritis damage to joint's cartilage
 - Jaw injury
 - Jaw muscle fatigue from grinding or clenching of teeth



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Lecture Outline

11. Temporomandibular joint (TMJ) disorders
 - a. Where the posterior condyle of the mandible articulates with the temporal bone
 - b. Causes include:
 - i. Arthritis damage to the joint's cartilage
 - ii. Jaw injury
 - iii. Jaw muscle fatigue from grinding or clenching of the teeth

TMJ Disorders: Assessment

- Symptoms may include:
 - Headache
 - Jaw pain
 - Aching around ear
 - Uneven or painful bite
 - Difficulty chewing
 - Locking of the joint

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Lecture Outline

c. Assessment and management

i. Symptoms include:

- (a) Headache
- (b) Jaw pain
- (c) Aching around the ear
- (d) An uneven bite and/or painful bite
- (e) Difficulty chewing
- (f) Locking of the joint causing difficulty either opening or closing the mouth

TMJ Disorders: Management

- Pain medication for symptoms
- Surgical interventions
- Managed by physician or dentist

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Lecture Outline

- ii. Usually managed by the patient's physician or dentist